

COMPLETE BEGINNER'S STUDY GUIDE

Content-Based Image Retrieval and Feature Extraction: A Comprehensive Review

Latif et al. (2019) - Mathematical Problems in Engineering
Explained for Literature Survey Beginners

0. What Is This Paper About? (The Big Picture)

Before diving into technical details, let us understand WHY this paper was written and WHAT problem it tries to solve.

0.1 The Core Problem: Finding Images

Imagine you have billions of images on the internet (like Google Images, Instagram, or Facebook). Now you want to find a specific image — say, a photo of a tiger. How do you find it?

Old way (Text-Based Search): You type the word 'tiger' and the system looks for images that have been manually labeled/tagged with the word 'tiger'. Problem: Someone has to manually write captions or tags for every single image. That is impossible for millions of images. Also, different people describe the same image differently.

New way (Content-Based Image Retrieval - CBIR): Instead of words, you give the system an IMAGE as input (like a photo of a tiger), and it automatically finds similar-looking images by analyzing the visual content itself — colors, shapes, textures, etc.

Real-world analogy: Think of CBIR like Google Lens on your phone. You take a photo of a flower and it finds similar flowers — without you typing anything. That is CBIR in action!

0.2 Why Is This Challenging?

The main challenge is called the 'Semantic Gap'. This means: Computers see images as numbers (pixels), but humans understand images by meaning. For example:

- A computer sees: millions of numbers representing color values
- A human sees: 'a tiger sitting under a tree in a forest'
- Bridging this gap between pixel numbers and human understanding is the core challenge of CBIR

0.3 What Does This Paper Do?

This paper is a REVIEW paper — meaning it does NOT propose one new method. Instead, it surveys and summarizes over 100 research papers from the last 12-15 years about CBIR and image feature extraction. Think of it as a 'textbook chapter' that explains the entire field.

For your literature survey: This paper is extremely valuable because it covers ALL the major approaches in CBIR: color features, texture features, shape features, spatial features,

machine learning, and deep learning. Understanding this paper = understanding the entire CBIR field.

1. About the Paper (Page 1 - Title & Abstract)

1.1 Authors and Institution

This paper was published in 2019 in the journal 'Mathematical Problems in Engineering' by researchers from:

- Mirpur University of Science and Technology (MUST), Pakistan — main institution
- RIPHAH International University, Islamabad, Pakistan
- Government College University, Faisalabad, Pakistan
- National Textile University, Faisalabad, Pakistan

1.2 Understanding the Abstract (Page 1)

The abstract (the short summary at the top of the paper) says the following key things — let us decode them:

What the Paper Says	What It Means in Simple English
Multimedia content analysis is applied in different real-world computer vision applications	Computers are being used to understand images, videos, and audio in the real world
More than millions of images are uploaded daily at Twitter, Facebook, Instagram	The internet is flooded with images every day — finding the right one is hard
Most search engines rely on text-based approaches using captions and metadata	Google Images mostly works by reading the title/tags of images, not the image itself
Significant gap between image feature representation and human visual understanding	Computers describe images differently than humans do — this is the 'semantic gap'
CBIR is focused to reduce the semantic gap	The goal of CBIR research is to make computers understand images more like humans do
We analyzed from low-level feature extraction to recent semantic deep-learning approaches	This paper covers everything from basic color analysis to modern AI (neural networks)

1.3 The 5 Research Questions This Paper Addresses

The paper explicitly states 5 objectives. These are important for your literature survey:

- How can CBIR performance be enhanced using low-level visual features (color, texture, shape)?
- How can the semantic gap between low-level features and high-level semantics be reduced?
- How important is spatial layout (where objects are in the image) for image retrieval?
- How can machine learning improve CBIR performance?
- How can deep neural networks (DNN) enhance learning from images?

2. Introduction — The Foundation (Pages 1-2)

2.1 How Traditional Image Search Works (and Why It Fails)

The introduction explains the two main approaches to image retrieval. Understanding the difference is crucial:

Approach 1: Text-Based Image Retrieval (Old Method)

- User types keywords (e.g., 'red sunset over ocean')
- System matches keywords to image tags/captions stored in database
- Problem 1: Someone must manually label millions of images — impossible at scale
- Problem 2: Different people use different words for same image ('car' vs 'automobile' vs 'vehicle')
- Problem 3: Returned images may not be visually relevant at all

Approach 2: Automatic Image Annotation

- System tries to automatically label images based on their visual content
- Still relies on detecting colors, edges, textures — but does it automatically
- Problem: Errors in automatic annotation can mislead the retrieval process

Approach 3: Content-Based Image Retrieval (CBIR) — The Focus of This Paper

- User provides a QUERY IMAGE as input (instead of text)
- System analyzes visual features of query image: color, shape, texture, spatial layout
- Compares these features with all images in the database
- Returns images that are visually similar to the query
- No manual labeling needed!

Key terms introduced: QBIC (Query-By-Image Content) and SIMPLicity are early successful CBIR systems mentioned in the paper. They were the first to demonstrate that visual features could be used for image search.

2.2 What Are 'Feature Vectors'? (A Critical Concept)

This term appears throughout the paper. Here is what it means:

An image is converted into a LIST OF NUMBERS called a feature vector. For example:

- Color feature vector: [30% red, 45% blue, 25% green, ...] — describes color distribution

- Texture feature vector: [roughness=0.7, regularity=0.4, coarseness=0.6, ...] — describes surface patterns
- Shape feature vector: [roundness=0.9, edges=120, corners=8, ...] — describes object shapes

When we want to compare two images, we compare their feature vectors using mathematical distance measures. Images with similar feature vectors are considered visually similar.

Simple analogy: A feature vector is like a recipe card for an image. Just as two recipes with similar ingredients produce similar dishes, two images with similar feature vectors look similar to each other.

2.3 Figure 1 Explained — Types of Image Queries (Page 3)

Figure 1 in the paper shows 4 different ways a user can search for images in a CBIR system. This is very important to understand:

Query Type	How It Works	Example
Query by Keyword	User types a text word like 'Tiger'	Type 'balloon' → system finds red balloon images
Query by Image	User submits an actual photo as query	Submit a photo of balloons → finds similar balloon photos
Query by Sketch	User draws a rough outline/sketch	Draw a stick figure → finds images of people
Query by Concept Layout	User specifies objects + their positions	Say 'horse in front, forest behind' → finds matching scenes

The figure also shows the concept of 'Abstract Thought' — the user's mental idea — being translated into a machine-readable query. This translation is where the semantic gap exists.

3. Color Features — Section 2 of Paper (Pages 2-5)

3.1 Why Color is Important

Color is one of the most basic and powerful visual features. The paper explains:

- Humans distinguish objects by color first — before shape or texture
- Color features are robust: they don't change much when image is rotated, scaled, or translated
- Color features are computationally cheap (fast to compute)

3.2 Key Color Feature Methods Discussed

A) Dominant Color Descriptor (DCD) — Shao et al.

Instead of storing ALL color information in an image, DCD selects only the 8 MOST DOMINANT colors. Like a painter's palette — instead of millions of shades, you pick the 8 most important ones.

- DCD is part of the MPEG-7 standard (an international standard for multimedia description)
- Uses histogram intersection algorithm to measure similarity between images
- Result: 89.64% accuracy on Corel dataset

B) Color Moment Invariant — Duanmu

Uses mathematical 'moments' (statistical measures) to describe color distribution. Key property: INVARIANT — meaning it produces the same result even if the image is slightly changed.

- Achieved 98.5% accuracy on COIL-100 dataset — one of the highest for color features

C) Pseudo-Zernike Moment Invariants — Zhang et al.

Zernike and pseudo-Zernike are mathematical polynomial functions used to represent images. Very technical but important concept:

- These functions are ORTHOGONAL — meaning each descriptor captures unique, non-overlapping information
- ROTATION INVARIANT — the descriptor remains the same even if you rotate the image
- Pseudo-Zernike is more robust to image noise than regular Zernike moments

D) Error Diffusion Block Truncation Coding (EDBTC) — Guo et al.

A two-step approach that both COMPRESSES images AND creates feature descriptors simultaneously:

- Step 1: Compress image using EDBTC (reduces storage space)

- Step 2: Extract two features from compressed form: Color Histogram Feature (CHF) and Bit Pattern Histogram Feature (BHF)
- CHF describes overall color; BHF describes the spatial arrangement of colors
- Result: 79.7% accuracy on Corel dataset

E) Multiscale Distance Coherence Vector (MDCV) — Jiexian et al.

This method addresses a problem: the same descriptor might describe different shapes. MDCV uses Gaussian functions to capture shape contours at multiple scales (zoom levels).

- INVARIANT to translation (shifting), rotation, and scaling
- 97% accuracy on MPEG-7 image database

3.3 Table 1 Summary (Page 5) — Color Features Comparison

The paper presents Table 1 summarizing all color feature methods. Key takeaways:

Method	Dataset Used	Accuracy	Key Advantage
Color Moment Invariant (Duanmu)	COIL-100	0.985 (98.5%)	Highest accuracy among color methods
Dominant Color Descriptor (Shao)	Corel	0.8964 (89.64%)	Compact representation, MPEG-7 standard
Color + Texture Fusion (Wang)	Corel	0.613 (61.3%)	Flexible estimation of human visual processing
EDBTC Features (Guo)	Corel	0.797 (79.7%)	Good for both compression and indexing
Vector Quantization (Islam)	Corel	0.9767 (97.67%)	Automatic categorization of image regions
MDCV (Jiexian)	MPEG-7	0.97 (97%)	Invariant to rotation, translation, scaling

Limitation noted in paper: Color moments alone are not sufficient — they cannot represent all regions of an image equally. Histogram-based features are computationally expensive. DCD is better for region-based retrieval.

4. Texture Features — Section 3 of Paper (Pages 4-5)

4.1 What is Texture?

Texture refers to the repeating patterns on the surface of objects. Examples:

- Smooth texture: polished metal, still water, skin
- Rough texture: stone wall, tree bark, sand
- Regular texture: brick wall, fabric weave, chessboard
- Irregular texture: clouds, grass, pebbles

Texture features are MORE SEMANTICALLY MEANINGFUL than color features because they represent groups of pixels (patterns) rather than individual pixel colors. However, they are sensitive to image noise.

4.2 Key Texture Feature Methods

A) Wavelet Moments (WMs) — Papakostas et al.

Wavelets are mathematical functions that decompose an image into different frequency components — like breaking music into bass, mid, and treble. Two types of wavelets were tested:

- WMs-1: Uses cubic B-spline wavelet (smooth, good for gradual texture changes)
- WMs-2: Uses Mexican hat wavelet (sharp, good for detecting edges and fine textures)
- Tested on 4 datasets: COIL, ORL, JAFFE, TRIESCH I
- Feature selection (keeping only the best features) greatly improved classification

B) Micro-Structure Descriptor (MSD) — Liu et al.

MSD captures the micro-structure of images (tiny local patterns). Tested on large Corel datasets:

- Corel-5000: 5,000 images across many categories
- Corel-10000: 10,000 images
- Uses HSV, RGB, and Lab color spaces together with texture orientation
- Best performance on HSV and Lab color spaces; poor on RGB
- Average precision: 55.92% on Corel-5000, 41.44% on Corel-10000

C) Color Co-occurrence Matrix — Wang et al.

The gray-level co-occurrence matrix (GLCM) is a classic texture descriptor that measures how often pairs of pixel values occur together. This paper improves it by adding COLOR information:

- Traditional: only considers gray-level pixel relationships
- Improved: considers color + gray-level relationships simultaneously

- Result: better precision than traditional GLCM because color adds extra discriminating information

D) SVM-Based Architecture — Irtaza and Jaffar

This approach uses Support Vector Machines (SVM) — a machine learning classifier — to categorize images by texture:

- Dataset: Corel image gallery — 1000 images (Corel A) + 9900 images (Corel B)
- SVM creates a mathematical boundary between image categories
- Shows consistent results at different numbers of returned images

Figure 2 Explained (Page 5): Figure 2 shows an SVM (Support Vector Machine) classification diagram. The dots represent images plotted in feature space. The line in the middle is the 'decision boundary' — images on one side are class +1, the other side class -1. The 'support vectors' are the closest dots to this boundary — they define where the line is drawn.

E) Local Derivative Radial Pattern (LDRP) — Fadaei et al.

LDRP is a texture descriptor that captures local derivative patterns — essentially how pixel values change in all directions around each pixel:

- Tested on Brodatz (112 grayscale images) and Vistex (54 color images) datasets
- Feature vector dimension: 3124 (high-dimensional)
- Faster at feature EXTRACTION but slower at feature MATCHING due to high dimensions
- Average precision: 80.81% on Brodatz, 91.91% on Vistex

4.3 Table 2 Summary — Texture Features (Page 6)

Key insight from the texture feature comparison:

- Texture features are MORE meaningful than color alone (they capture surface patterns)
- Main weakness: texture features are SENSITIVE TO NOISE — small amounts of image noise can significantly change texture descriptors
- Texture description depends on object shapes — the same texture looks different on curved vs. flat surfaces

5. Shape Features — Section 4 of Paper (Page 5 & Figure 3)

5.1 What Are Shape Features?

Shape is arguably the most intuitive visual feature — humans recognize objects primarily by their shape. Shape features describe the geometric form of objects in images.

5.2 Figure 3 Explained — The Shape Feature Tree (Page 7)

Figure 3 is one of the most information-rich figures in the paper. It shows a hierarchical classification of ALL shape representation methods. Let us understand the tree structure:

Level 1: Shape Representation (The Root)

All methods divide into two main branches: Contour-based (uses boundary/outline) and Region-based (uses interior area).

Branch 1: One-Dimensional Function for Shape Representation

- Complex coordinate: Represents boundary as complex numbers ($x + iy$)
- Centroid distance function: Distance from center of shape to each boundary point
- Tangent angle: Angle of the tangent line at each boundary point
- Contour curvature: How much the boundary curves at each point
- Area function: Area swept as you trace the boundary
- Triangle-area representation: Area of triangles formed by boundary points
- Chord length function: Distances between pairs of boundary points

Branch 2: Polygonal Approximation

- Merging methods: Combine small line segments into larger ones
- Splitting method: Divide curves at points of high curvature

Branch 3: Spatial Interrelation Feature

- Adaptive grid resolution / Boundary box / Convex hull / Chain code: Different ways to encode the boundary
- Smooth curve decomposition: Breaking complex shapes into simple curve segments
- Shape matrix / Shape context / Chord distribution: Statistical descriptions of shape
- Shock graph: Represents shape using its 'skeleton' (like the bones inside a shape)

Branch 4: Moments

- Boundary moment: Statistical moments computed from the boundary
- Region moments: Statistical moments computed from the filled region

Branch 5: Scale Space Methods

- Curvature scale space (CSS): Tracks how boundary curves change at different smoothing levels — very popular for shape retrieval
- Intersection point map: Tracks where horizontal lines cross the boundary

Branch 6: Shape Transform Domain

- Fourier descriptors: Represent boundary as sum of sine waves — rotation invariant
- Wavelet transform: Multi-scale shape analysis
- Angular radial transformation: Used in MPEG-7 shape descriptor
- R-transform / Shapelet descriptors: Other mathematical representations

Application example: Shape features are especially important in trademark/logo retrieval (cited in the paper). When checking if a new logo is too similar to an existing trademark, shape comparison is critical.

6. Spatial Features — Section 5 of Paper (Pages 5-6)

6.1 Why Spatial Information Matters

Consider two images: one with a dog on the LEFT and a cat on the RIGHT, another with a cat on the LEFT and a dog on the RIGHT. If you only look at what objects exist, these images seem identical. But spatially, they are different! Spatial features capture WHERE objects appear in the image.

6.2 Bag of Visual Words (BoVW) — The Foundation

BoVW is one of the most important models in CBIR. Understanding it is crucial:

- Step 1: Extract local features (patches) from all training images
- Step 2: Cluster these patches into groups — each group is a 'visual word'
- Step 3: Represent each image as a histogram of how many times each visual word appears
- Problem: BoVW IGNORES spatial layout — it only counts words, not where they appear

Text analogy: BoVW is like summarizing a book by counting how often each word appears, without caring about the order or position of words. You know the book contains 'murder', 'mansion', and 'butler' but not whether the butler murdered someone in the mansion.

6.3 Spatial Pyramid Matching (SPM)

SPM (by Lazebnik et al., cited as [43] in paper) is a popular fix for BoVW's spatial blindness:

- Divides the image into a pyramid of regions (whole image, then 4 quadrants, then 16 sub-regions, etc.)
- Computes BoVW histogram for each region separately
- Concatenates all histograms — this captures rough spatial location
- Limitation: SPM is insensitive to scaling and rotations

6.4 Advanced Spatial Methods by the Research Group

The paper heavily cites work by the authors' own research group (Zafar, Ali, Khan, et al.):

HGSIR — Hybrid Geometric Spatial Image Representation (Ali et al.)

- Computes histograms for rectangular, triangular, AND circular areas of the image
- Combines multiple geometric shapes to capture spatial distribution comprehensively
- Outperforms state-of-the-art methods on 5 datasets

PIW — Pairs of Identical Visual Words (Khan et al.)

- Focuses on RELATIONSHIPS between similar visual words in the image
- Computes histogram of orientations of line segments formed by pairs of same visual words
- Advantage: adds global spatial information without much extra computation

Circular Tilings with Rotation-Invariant Angles (Anwar et al.)

- Adds spatial information using circular grid divisions (like a dartboard pattern)
- Modifies angle histograms to be ROTATION INVARIANT
- Tested on coin image classification (where rotation is very common)

Orthogonal Vector-Based Spatial Representation (Ali et al.)

- Computes orthogonal (perpendicular) vectors between pairs of identical visual words
- The size of orthogonal vector encodes position information
- Designed specifically for satellite image classification where rotation invariance is critical

7. Low-Level Feature Fusion — Section 6 of Paper (Pages 6-9)

7.1 Why Fusion Works Better

The key insight: NO SINGLE FEATURE TYPE captures everything about an image. Combining multiple features (fusion) gives better results. Think of it like:

- Color alone: Can find images with same colors but completely different content
- Texture alone: Can find images with same surface patterns but different objects
- Shape alone: Can find images with same outlines but different colors
- Color + Texture + Shape together: Much more discriminating and accurate

7.2 Color + DWT Fusion — Ashraf et al. [55]

This paper combines color features with Discrete Wavelet Transform (DWT) for texture:

- Color spaces used: RGB (Red-Green-Blue) and YCbCr
- Why YCbCr? The human visual system separates brightness (Y) from color (Cb, Cr) — YCbCr mimics this separation, solving color vibration problems
- Edge detection: Canny edge detector extracts shape information
- Haar Wavelet Transform applied to enhance search efficiency
- Final step: Artificial Neural Network (ANN) applied for retrieval
- Result: Outperforms other existing CBIR systems

7.3 MDCBIR-MF — Ashraf et al. [56]

A more advanced fusion approach combining:

- Color Moments for color description
- Discrete Wavelet Transform (DWT) for texture
- Gabor Wavelet for additional texture detail
- Color and Edge Directory Descriptor for shape
- Result: 87.5% precision on Corel-1000 dataset

7.4 Hybrid Features with Distance Metrics — Mistry et al. [57]

This paper uses 4 different feature extractors combined:

- BSIF (Binarized Statistical Image Features): Extracts binary features from grayscale patches
- CEDD (Color and Edge Directivity Descriptor): Captures both color and edge direction
- HSV Color Histogram: Describes color distribution in Hue-Saturation-Value space
- Color Moment: Statistical description of color components

- Result: 87.5% precision, significantly outperforms existing methods

7.5 Image Feature Information Fusion — Ahmed et al. [58]

Fuses spatial color features + shape features (edges and corners):

- RGB for color extraction
- Gray-level image for edge and corner detection (shape features)
- Key insight: Edges + corners = powerful shape descriptor
- Dimension reduction applied (PCA-like) to reduce feature vector size
- Bag of Words (BoW) used for quick indexing
- Result: 90% precision for Africa and building image categories

7.6 LBP + Color Information Feature — Liu et al. [59]

LBP (Local Binary Pattern) is a classic texture descriptor. For each pixel:

- Step 1: Compare pixel with its 8 neighbors
- Step 2: Assign 1 if neighbor $\geq$ center pixel, 0 otherwise
- Step 3: Read the 8 bits as a binary number — this is the LBP code
- Problem: LBP alone doesn't describe color well
- Solution: Combine LBP texture features with CIF (Color Information Feature)
- Result: 84.1% on Brodatz, 95.2% on Vistex datasets

7.7 Gaussian Copula + Gabor Wavelets — Li et al. [61]

An advanced statistical approach:

- Gabor filter: Detects oriented edges and textures at multiple scales and orientations
- Gabor wavelets decompose color image into subbands
- Three types of dependencies captured: directional, color, and scale
- Gaussian copula model captures statistical dependencies between subbands
- Four Kullback-Leibler distances (KLD) measure similarity between images
- Tested on ALOT and STex texture datasets — outperforms state-of-the-art

7.8 Table 3 — Fusion Feature Summary (Page 9)

Paper	Features Combined	Dataset	Precision
Nazir et al.	HSV Color Histogram + DWT + Edge Histogram	Corel-1K	73.5%
Ashraf et al. [56]	Color Moments + DWT + Gabor + Edge	Corel-1K	87.5%

Paper	Features Combined	Dataset	Precision
Mistry et al.	BSIF + CEDD + HSV Histogram + Color Moment	Wang Dataset	87.5%
Ahmed et al.	Spatial Color + Shape (edges+corners) + BoW	Corel-1K	90% (Africa/Building)
Liu et al.	LBP (texture) + CIF (color)	Brodatz/Vistex	84.1% / 95.2%

8. Local Feature-Based Approaches — Section 7 (Pages 9-13)

8.1 What Are Local Features?

Instead of describing the ENTIRE image with one feature vector (global features), local features describe small PATCHES or KEYPOINTS of the image. Benefits:

- More robust to occlusion (when part of image is blocked)
- More robust to viewpoint changes
- Can match specific parts of images (e.g., find the same face in different photos)

8.2 Sparse Representation for Image Similarity — Kang et al. [64]

Sparse representation means: represent an image as a combination of only a FEW items from a large dictionary. Like describing a color using only 3 primary colors — you need only 3 numbers instead of millions.

- Learns a dictionary of feature patches from training images
- Represents each image sparsely (using only a few dictionary elements)
- Applied to: image copy-move detection, retrieval, and recognition
- Tested on Corel-1000, COIL-20, COIL-100, Caltech-101
- 98.5% accuracy on COIL-20 dataset

8.3 Sketch-Based Image Retrieval (SBIR) — Li et al. [71]

What if you cannot find an example image to query with? You draw a sketch instead!

- SBIR uses edges/contours from sketches to find real photos
- Challenge: sketch and photo look very different but describe the same object
- Method: Product Quantization + Sparse Coding for efficient codebook construction
- Result: 98% accuracy on Eitz benchmark — one of the highest SBIR results

8.4 Figures 5, 6, 7 — Benchmark Datasets (Pages 12-13)

These figures show sample images from three commonly used datasets. Understanding these datasets is ESSENTIAL for your literature survey:

Dataset (Figure)	Size	Content	Typical Use
Corel-1500 (Fig 5)	1,500 images, 15 classes	Natural scenes: animals, scenery, people, food	Standard CBIR benchmark — used in most papers
Caltech-101 (Fig 6)	9,146 images,	Objects: motorcycles, faces,	Object recognition and

Dataset (Figure)	Size	Content	Typical Use
	101 classes	watches, animals	classification
15Scene (Fig 7)	~4,500 images, 15 scene classes	Grayscale: coast, mountain, office, street, bedroom	Scene classification benchmark

8.5 Figure 8 — Machine Learning Techniques for CBIR (Page 13)

Figure 8 shows the main machine learning algorithms used in CBIR. Here is what each one does:

ML Technique	How It Works (Simple Explanation)	Use in CBIR
Support Vector Machine (SVM)	Finds the best mathematical boundary between classes in feature space	Image classification, texture categorization
Artificial Neural Network (ANN)	Loosely models how neurons in the brain work; learns patterns from examples	Feature learning, image annotation
Decision Tree	Asks a series of yes/no questions to classify images	Region-based image retrieval
Bayesian Method	Uses probability theory to classify; updates probabilities as it sees more data	Image annotation, probabilistic retrieval
Nonparametric Approach	Makes no assumptions about data distribution; examples: KNN (K-Nearest Neighbors)	Direct image similarity matching
Parametric Approach	Assumes data follows a specific distribution (e.g., Gaussian); estimates parameters	Statistical image modeling

8.6 Figure 9 — Machine-Human Interactions (Page 13)

Figure 9 shows a diamond diagram of how machine capabilities and human capabilities meet in the middle through 'human-centered computing'. The key idea:

- Machines are good at: statistical analysis, data mining, data management, compression
- Humans are good at: cognition, perception, visual intelligence, decision-making
- The goal of CBIR is to find 'the best of both sides' through semantic-based approaches
- This intersection — where machines understand images like humans — is the ultimate goal

9. CBIR Using Deep Learning — Section 8 (Pages 14-16)

9.1 Why Deep Learning Changed Everything

Deep learning, especially Convolutional Neural Networks (CNN), revolutionized image analysis after 2012. Here is why:

- Traditional methods: Humans DESIGN features (color histograms, LBP, etc.) — called 'handcrafted features'
- Deep learning: The network LEARNS features automatically from millions of examples
- Result: Deep learning features often outperform handcrafted features on large datasets

Key reference: Krizhevsky et al. [87] won the ImageNet competition in 2012 with AlexNet CNN — achieving 37.5% top-1 and 17.0% top-5 error rate. This moment marked the beginning of the 'deep learning era' in computer vision.

9.2 Figure 10 — CNN Architecture Explained (Page 14)

Figure 10 shows a CNN (Convolutional Neural Network) architecture for image classification. Let us decode each layer:

Layer	Size in Figure	What It Does
L0 (Input)	512 x 512 pixels	The raw input image
L1 (Conv Layer 1)	256 x 256	First convolution: detects basic edges and simple patterns
L2 (Conv Layer 2)	128 x 128	Second convolution: detects combinations of edges (corners, curves)
L3 (Conv Layer 3)	64 x 64	Third convolution: detects parts of objects (eyes, wheels, etc.)
L4 (Conv Layer 4)	32 x 32	Fourth convolution: detects whole objects or complex patterns
F5 (Fully Connected)	1D vector	Combines all spatial features into a global feature vector
F6 (Output)	Class probabilities	Final classification: probability of each category

Each convolution layer learns to detect increasingly complex features — from edges to objects. This hierarchy is why CNNs are so powerful.

9.3 TF-IDF with CNN — Kondylidis et al. [82]

TF-IDF (Term Frequency-Inverse Document Frequency) is a concept from text retrieval applied to images:

- TF (Term Frequency): How often a visual word (CNN filter activation) appears in THIS image
- IDF (Inverse Document Frequency): How rare this visual word is across ALL images in database
- Visual word that appears often in one image but rarely elsewhere = highly descriptive
- Result: Improved image retrieval on 4 datasets including Oxford 5k and Paris 6k benchmarks

9.4 Deep Hashing for Efficient Retrieval

Hashing converts feature vectors into short binary codes (e.g., 32 bits). Benefits: fast search, small storage. Three papers cover this:

Supervised Deep Hashing (SSDH) — Yang et al. [86]

- Constructs hash function from a 'latent layer' in the neural network
- Unifies retrieval AND classification in a single model
- Binary codes learned to preserve semantic similarity
- Simple to implement by modifying existing classification networks

Pairwise Deep Ranking Hashing (PDRH) — Shi et al. [83]

- Applied to histopathology (medical tissue) images
- 97.94% classification accuracy on 5,356 skeletal muscle + 2,176 lung cancer images
- Models pairwise relationships (which images are similar/dissimilar) in objective function

Semantics Assisted Visual Hashing (SAVH) — Zhu et al. [84]

- UNSUPERVISED method — does not need labeled training data
- Two components: offline learning (builds hash function) and online learning (applies it)
- Uses topic hypergraph to extract semantic information from text associated with images

9.5 Bilinear CNN — Alzu'bi et al. [85]

Instead of one CNN, use TWO parallel CNNs to extract features simultaneously:

- Each CNN model captures different aspects of image content
- Features extracted from intermediate convolutional layers (not just the last layer)
- Advantage: Better at learning complex images with multiple semantics
- Speed: Only 10 milliseconds to extract features and search database

- Very small disk size needed for image representation

9.6 Table 5 — Deep Learning CBIR Summary (Page 16)

Authors	Purpose	Model	Key Result
Krizhevsky et al.	Image Classification	AlexNet CNN	37.5% top-1 error on ImageNet (breakthrough)
Sun et al.	Face Verification	DeepID (ConvNets)	97.45% accuracy on LFW dataset
Shi et al.	Medical Image Retrieval	PDRH Deep Hashing	97.49% classification accuracy
Kondylidis et al.	Content-Based Retrieval	CNN + TF-IDF	Improved results on Oxford/Paris datasets
Li et al.	Social Image Understanding	Deep Collaborative Embedding	0.512 on MIRFlickr, 0.632 on NUS-WIDE

10. Feature Extraction for Face Recognition — Section 9 (Pages 15-17)

10.1 Why Face Recognition Gets Its Own Section

Face recognition is one of the most important applications of feature extraction. The paper dedicates a full section to it because:

- Face images are complex — lighting, age, pose, expression all change appearance
- It requires a combination of local and global features
- Deep learning has dramatically improved face recognition performance

10.2 3D Face Recognition Methods (Ratyal et al.)

Regular (2D) photos lose depth information. 3D face recognition uses 3D scans (like depth cameras):

Method 1 — PCA + Mahalanobis Distance [94]

- ICS (Intrinsic Coordinate System) used to correct face pose/alignment
- PCA (Principal Component Analysis): reduces feature dimensions while keeping most information
- MahCos distance: combines Mahalanobis and Cosine distance for classification
- Re-ranking via weighted Borda count method
- Tested on GavabDB and FRGC v2.0 datasets

Method 2 — Multi-view 3D Face Recognition [95]

- Two-pass alignment handles both frontal AND profile face images
- PCA-based features with multistage unified classifier and SVM
- Tested on 4 benchmarks: GavabDB, Bosphorus, UMB-DB, FRGC v2.0

Method 3 — Deep Learning for 3D Faces [97]

- Uses AlexNet (deep CNN) for feature extraction from 3D face data
- Coarse-to-fine alignment algorithm for precise face normalization
- Both AlexNet features and SVM classification tested separately

10.3 Age-Invariant Face Recognition (Sajid et al.)

How do you recognize the same person's face 20 years apart? This is the age-invariant face recognition problem:

- Hybrid generative-discriminative model: combines GAN-generated features with discriminative learning
- Demographic traits (age group, gender, race) used to enhance recognition
- Tested on MORPH and FG-Net aging datasets

10.4 Facial Asymmetry-Based Methods (Sajid et al.)

Human faces are not perfectly symmetric — left and right halves differ slightly. These differences encode useful information:

- Facial asymmetry used to estimate gender and ethnicity
- Asymmetric features used to grade facial palsy (nerve disease) severity using GAN
- Asymmetric right face images better for age estimation than left face images
- Data augmentation with celebrity makeup styles to recognize faces across makeup variations

11. Distance Measures — Section 10 (Page 17)

11.1 What Are Distance Measures?

Once we have feature vectors for two images, we need to measure HOW SIMILAR they are. This is done using distance measures. Smaller distance = more similar images.

Distance Measure	Formula Concept	Best Used For
Euclidean Distance	Square root of sum of squared differences	Continuous, normally distributed features
Manhattan Distance (L1)	Sum of absolute differences	Robust to outliers; good for sparse features
Minkowski Distance	Generalization of Euclidean and Manhattan	Adjustable with parameter p
Cosine Similarity	Angle between two feature vectors (not magnitude)	Text-like features, BoVW histograms
Histogram Intersection	Minimum overlap between two histograms	Color and texture histograms
Chi-Square Distance	Weighted difference between histogram bins	Histogram comparison, especially color
Mahalanobis Distance	Euclidean distance adjusted for feature correlations	When features are correlated
KL Divergence	Information-theoretic difference between distributions	Probabilistic/statistical models (Gabor, GMM)

Key insight from paper: The choice of distance measure significantly affects retrieval quality. There is no single best measure — it depends on the type of feature vector and the application domain.

11.2 Figure 11 — Image Retrieval Results (Page 18)

Figure 11 shows visual results of a CBIR system doing top-5 to top-25 image retrieval. Each row shows:

- Leftmost image: the QUERY image submitted by the user
- Arrow pointing right: system searches the database
- Remaining images: the top-N retrieved results ordered by similarity
- Good results: all retrieved images are visually similar to the query

The figure demonstrates 5 different queries (rows) showing the system's retrieval capability.

12. Performance Evaluation Criteria — Section 11 (Pages 17-18)

12.1 Why Evaluation Metrics Matter

How do we know if a CBIR system is GOOD? We need standardized metrics to compare different systems fairly. This section explains the main evaluation metrics — you MUST understand these for your literature survey.

12.2 Precision (P)

Precision answers: 'Of all the images the system retrieved, what fraction are actually relevant?'

Formula: Precision = True Positives / (True Positives + False Positives) = (relevant images retrieved) / (total images retrieved)

Example: You query for 'dogs'. System returns 20 images. 15 are dogs, 5 are cats. Precision = $15/20 = 0.75$ (75%)

12.3 Recall (R)

Recall answers: 'Of all the relevant images in the database, what fraction did the system find?'

Formula: Recall = True Positives / (True Positives + False Negatives) = (relevant images retrieved) / (total relevant images in database)

Example: Database has 100 dog images total. Your query retrieved 15 of them. Recall = $15/100 = 0.15$ (15%)

12.4 The Precision-Recall Trade-off

There is always a trade-off between precision and recall:

- If you retrieve FEW images: High precision (everything you retrieve is relevant) but LOW recall (you missed many relevant images)
- If you retrieve MANY images: Low precision (many irrelevant images) but HIGH recall (you found most relevant images)
- The P-R curve shows this trade-off at different retrieval thresholds

12.5 F-Measure

Formula: $F = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$ — This is the harmonic mean of P and R. Higher F-measure = better overall performance.

12.6 Average Precision (AP)

AP summarizes the entire precision-recall curve into a single number:

Formula: $AP = \text{Sum of } [P(k) \times R(k)] \text{ for each relevant image } k, \text{ divided by total number of relevant images } NRI$

It rewards systems that rank relevant images higher in the result list.

12.7 Mean Average Precision (MAP)

MAP averages the AP scores across MULTIPLE different queries:

Formula: $MAP = \text{Sum of } AP(q) \text{ for all queries } q, \text{ divided by total number of queries } S$

- This is the MOST COMMONLY reported metric in CBIR papers
- Higher MAP = system works well across diverse query types

12.8 ANMRR — Averaged Normalized Modified Retrieval Rank

- Used specifically for MPEG-7 color experiments
- Produces results in range $[0, 1]$ — SMALLER values mean BETTER performance
- Unlike precision/recall, ANMRR = 0 means perfect retrieval

12.9 MNRO — Mean Normalized Retrieval Order

- Proposed by Chatzichristofis et al. to represent system behavior without bias for top-k retrievals
- Useful for comparing systems that retrieve different numbers of results

13. Conclusion and Future Directions — Section 12 (Pages 18)

13.1 What the Paper Concludes

The paper summarizes 12-15 years of CBIR research and draws these key conclusions:

Finding 1: Single features are not enough

Due to diversity in image datasets and non-homogeneous image properties, no single feature type (color, texture, or shape alone) can adequately represent all images.

Finding 2: Feature fusion improves performance

Using low-level features in combination (fusion) significantly improves CBIR performance. Combining local and global features is one of the most promising directions.

Finding 3: Local features reduce semantic gap

Fusing different local features (which represent images as patches) reduces the semantic gap better than global features, because local features can capture meaningful sub-regions.

Finding 4: Deep learning outperforms handcrafted features

Recent deep neural networks outperform traditional handcrafted features on most datasets — but they require large labeled training sets and high computational resources.

13.2 Future Research Directions Identified

The paper identifies 4 key areas for future research — these are VERY IMPORTANT for your own research direction:

Future Direction	Current Problem	Proposed Solution Direction
Feature Dimension Optimization	High-dimensional feature vectors cause 'curse of dimensionality' and overfitting	Optimize feature representation to reduce dimensions while maintaining discriminability
Combining Local + Global Features	Local features miss global context; global features miss fine details	Novel fusion frameworks that intelligently combine both levels
Unsupervised Deep Learning on Large	Deep networks need millions of labeled images for	Evaluate deep networks in unsupervised learning mode on large unlabeled datasets

Future Direction	Current Problem	Proposed Solution Direction
Unlabeled Datasets	training — expensive to collect	
3D Face Recognition with Deep Features	3D face recognition still short of needed performance	Apply deep learning to 3D face data; explore new 3D face datasets (LAP-1, LAP-2)

14. Quick Reference — Datasets Used in the Paper

For your literature survey, you need to know the standard datasets. Here is a complete reference:

Dataset Name	Size	Type	Used For
Corel-1000 (Wang Dataset)	1,000 images / 10 classes (100 each)	Color photos: Africa, beach, buildings, buses, dinosaurs, elephants, flowers, horses, mountains, food	Standard CBIR benchmark — most commonly used
Corel-5000	5,000 images	Diverse natural and artificial scenes	Large-scale CBIR evaluation
Corel-10000	10,000 images	Diverse natural and artificial scenes	Large-scale CBIR evaluation
COIL-20	1,440 images / 20 objects	Objects photographed from 72 angles (360°)	Rotation invariance testing
COIL-100	7,200 images / 100 objects	100 objects, 72 views each	Object recognition, color features
Caltech-101	~9,000 images / 101 classes	Objects: animals, vehicles, faces, furniture	Object recognition benchmark
Caltech-256	30,607 images / 256 classes	More challenging than Caltech-101	Large-scale object recognition
15Scene	~4,500 images / 15 classes	Grayscale indoor/outdoor scenes	Scene classification
Brodatz	112 grayscale images	Various textures: leather, weave, brick, etc.	Texture-based retrieval
Vistex	54 color texture images (640-864 images)	Color textures: fabric, bark, water, etc.	Color texture retrieval
ALOT	27,750 images / 250 textures	Material surface textures	Texture image retrieval
LFW (Labeled Faces in Wild)	13,000+ face images / 5,749 people	Real-world face photos in diverse conditions	Face recognition benchmark
GavabDB	549 3D face scans / 61 subjects	3D facial scans with pose and expression variation	3D face recognition
FRGC v2.0	50,000 images / 466 subjects	2D + 3D face images	3D face recognition
MORPH	55,000 images / 13,000 subjects	Faces aged 16-77 years over time	Age-invariant face recognition
ImageCLEF	Medical images	X-rays, CT scans, MRI images	Medical image retrieval

15. Glossary of Key Terms

All technical terms used in this paper, explained simply:

Term	Simple Explanation
CBIR (Content-Based Image Retrieval)	Searching for images based on their visual content (colors, textures, shapes) rather than text labels
Semantic Gap	The difference between how computers see images (as numbers) and how humans understand images (by meaning)
Feature Vector	A list of numbers that describes an image's visual properties. Like a 'fingerprint' of the image
Feature Extraction	The process of converting an image into a feature vector
Low-level Features	Basic visual properties: color, texture, shape, spatial layout — computed directly from pixels
High-level Semantics	Human-understandable concepts: 'a dog', 'a sunset', 'a happy person' — hard for computers to detect automatically
Precision	Of all images retrieved, what fraction are relevant? (quality of results)
Recall	Of all relevant images, what fraction were retrieved? (completeness of results)
MAP (Mean Average Precision)	Average quality across many different queries — the gold standard metric in CBIR
Bag of Visual Words (BoVW)	Representing an image as a histogram of visual 'words' (local feature clusters), ignoring spatial layout
Codebook / Visual Vocabulary	A dictionary of visual words learned by clustering local image patches
SVM (Support Vector Machine)	A machine learning algorithm that finds the best separating boundary between classes
CNN (Convolutional Neural Network)	A deep learning architecture that automatically learns hierarchical image features
Convolution	Sliding a small filter over an image to detect local patterns (edges, textures, etc.)
Hashing	Converting a feature vector into a short binary code for fast similarity search
Local Features	Features extracted from small patches or keypoints of an image (e.g., SIFT, SURF, LBP)
Global Features	Features that describe the entire image as a whole (e.g., color histogram)
Feature Fusion	Combining multiple feature types to get a richer image description
DWT (Discrete Wavelet Transform)	A mathematical transformation that analyzes images at multiple frequency levels
Gabor Filter	An orientation-selective filter that detects texture patterns at specific angles and scales

Term	Simple Explanation
LBP (Local Binary Pattern)	Encodes local texture by comparing each pixel with its neighbors
PCA (Principal Component Analysis)	Reduces feature dimensionality by finding directions of maximum variance
GAN (Generative Adversarial Network)	A deep learning model where two networks compete: one generates fake images, one detects fakes
Sparse Representation	Representing an image using a SMALL number of elements from a large dictionary
SIFT (Scale-Invariant Feature Transform)	A local feature detector robust to scale and rotation changes
Transfer Learning	Using a CNN trained on one task (e.g., ImageNet) as a starting point for another task
Overfitting	When a model learns training data too well but performs poorly on new data
MPEG-7	International standard for multimedia content description — defines standard feature descriptors

16. Summary: Key Takeaways for Your Literature Survey

16.1 The Evolution of CBIR Approaches

The paper traces a clear evolution. For your literature survey, frame this timeline:

Era	Approach	Key Methods	Main Limitation
1990s-2000s	Low-level Single Features	Color histograms, Zernike moments, Gabor texture	Large semantic gap; single feature insufficient
2000s-2010s	Feature Fusion	Color + texture + shape combinations; BoVW model	High computational cost; still semantic gap
2010s	Machine Learning + Local Features	SVM, sparse representation, SIFT, spatial pyramids	Needs labeled data; feature engineering required
2015-Present	Deep Learning (CNN)	AlexNet, deep hashing, bilinear CNN, GAN-based	Needs huge labeled datasets; high compute cost

16.2 The 6 Most Important Concepts to Remember

- **Semantic Gap:** The fundamental problem CBIR aims to solve — bridging pixel-level data and human understanding
- **Feature Hierarchy:** Low-level (pixels) → Mid-level (patches, textures) → High-level (objects, scenes, semantics)
- **Fusion is Key:** No single feature type is sufficient — combining multiple features always improves performance
- **Spatial Layout Matters:** WHERE objects are in an image is as important as WHAT the objects are
- **Deep Learning Dominates:** CNNs automatically learn better features than handcrafted methods (given enough data)
- **Evaluation Standards:** Precision, Recall, MAP are the universal metrics — always understand what metric a paper uses

16.3 Citation-Worthy Claims from This Paper

When writing your literature survey, you can cite this paper for these specific claims:

- 'Millions of images are uploaded daily on social media, making image search a critical challenge [Latif et al., 2019]'
- 'CBIR systems use low-level visual features such as color, texture, shape, and spatial layout [Latif et al., 2019]'

- 'The semantic gap between low-level image representation and high-level human understanding is the core challenge of CBIR [Latif et al., 2019]'
- 'Deep neural networks outperform handcrafted features on large-scale image datasets but require significant computational resources [Latif et al., 2019]'
- 'Feature fusion of multiple low-level descriptors significantly improves CBIR performance compared to single-feature approaches [Latif et al., 2019]'

16.4 Research Gaps You Can Explore

Based on the future directions section, these are open research problems you could address:

- Reducing feature vector dimensionality without losing discriminative power
- Effective unsupervised or semi-supervised deep learning for image retrieval without massive labeled datasets
- Better fusion strategies that intelligently weight different feature types
- CBIR for specific domains: medical images, satellite images, fashion, art
- Combining sketch-based and content-based retrieval for multimodal queries
- Real-time CBIR on mobile devices with limited computation

— End of Study Guide —

Based on: Latif et al. (2019). Content-Based Image Retrieval and Feature Extraction: A Comprehensive Review. Mathematical Problems in Engineering, Article ID 9658350.